Swasthya Twin

Explainable, Adaptive Patient Digital Twin

Understand the change. Explore the possibilities. Act earlier.

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Problem Statement

Healthcare decisions often rely on single-point measurements or population averages that may not reflect an individual's unique health trajectory.

Key Challenges:

- Lack of longitudinal health tracking
- No tools for personal baseline comparison
- Limited ability to explore hypothetical interventions
- Opaque risk assessments without explanations

Our Solution:

Swasthya Twin creates a patient-specific virtual health model that maintains longitudinal observations, computes personalized risk assessments, and enables evidence-based scenario exploration.

Healthcare Use Case

Initial Clinical Focus: Cardiometabolic Health

Key Capabilities:

- Baseline establishment from individual observations
- Longitudinal trend tracking with rolling averages
- Personalized risk assessment with explainable contributions
- Hypothetical scenario simulation for intervention exploration
- Data quality assessment and uncertainty communication

Target Users:

- Researchers exploring Digital Twin concepts
- Healthcare technology developers
- Educational demonstrations of personalized health modeling

System Architecture

Modular, Layered Design

Presentation Layer: Landing page, Demo UI, API routes

Application Layer: Flask application, request handling, security

Service Layer: Digital Twin Engine, Prediction Engine, Scenario Engine, Validators

Repository Layer: Patient, Observation, Scenario repositories

Persistence Layer: SQLite database with schema and indexes

Technology Stack:

- Backend: Python 3.14.6, Flask 3.0.3, SQLite
- Frontend: HTML5, CSS3, Vanilla JavaScript, Jinja2
- Dependencies: NumPy, Pandas, Gunicorn

Digital Twin Engine

Core Functionality

Computes patient-specific health state from longitudinal observations

Key Computations:

- Baseline calculation: Average of first N observations (default: 3)
- Rolling averages: Time-weighted averages over configurable windows
- Change from baseline: Current value compared to personal normal
- Trend direction: Increasing, decreasing, or stable patterns
- Data quality assessment: Completeness, freshness, consistency scores

Supported Features:

- Resting Heart Rate, Blood Pressure (systolic/diastolic)
- Body Mass Index, Blood Glucose
- Physical Activity, Sleep Duration

Risk Assessment Model

Cardiometabolic Risk Estimator v1.0.0

Rules-based screening with personalization (not a trained ML model)

Methodology:

- Evidence-based clinical thresholds (ACC/AHA, WHO, ADA)
- Feature assessment against personalized baselines
- Weighted sum of feature contributions
- Confidence scoring based on data quality

Output:

- Risk Score: 0.0 to 1.0
- Risk Level: LOW, MODERATE, HIGH
- Feature Contributions: Per-feature analysis with explanations
- Confidence Score: Based on data quality

Limitations Documented:

- Not validated for clinical use
- Does not account for medications, comorbidities, family history

Scenario Simulation

Hypothetical Intervention Exploration

Users can modify health variables within reasonable bounds and see how hypothetical changes affect risk assessments without modifying historical data.

Workflow:

1. User creates scenario with feature modifications
2. System validates modifications are within safe ranges
3. Load baseline Digital Twin state and risk
4. Apply modifications and compute new risk
5. Compare baseline vs scenario with risk change quantification

Key Innovation:

Non-destructive testing enables evidence-based exploration of interventions with clear baseline-to-scenario comparison.

Key Innovations

Baseline-to-Scenario Evidence Chain

Every meaningful change shows a traceable chain: Observed data → validated features → comparison with personal baseline → algorithm output → explanation → hypothetical scenario comparison

Reproducible Calculations

- Fixed random seeds for synthetic data generation
- Deterministic state computations
- Versioned model parameters
- Stored calculation metadata

Explainable Risk Assessment

- Per-feature contribution analysis
- Evidence-based clinical thresholds
- Clear uncertainty communication
- Explicit limitations documentation

Security and Privacy

Implemented Measures

- Input validation for all user inputs
- Parameterized SQL queries (SQL injection prevention)
- Secure file upload handling (CSV only, 16 MB limit)
- CSRF protection for state-changing forms
- Environment-based secret keys
- No sensitive data in logs
- No external API calls
- Local-only data storage

Limitations

- Ephemeral storage on free deployment tier
- No authentication system (demo mode)
- No encryption at rest (SQLite)

Evaluation Results

Functional Testing

- ✓ Database initialization
- ✓ Patient CRUD operations
- ✓ Observation validation
- ✓ CSV import/export
- ✓ Digital Twin state computation
- ✓ Risk assessment
- ✓ Scenario simulation

Reproducibility Testing

- ✓ Synthetic data generation with fixed seed
- ✓ State computation determinism
- ✓ Risk assessment reproducibility
- ✓ Scenario consistency

Performance

- • Startup time: <2 seconds
- • Memory footprint: ~100-200 MB typical
- • API response time: <500ms for typical operations

Note: This prototype has not been validated on clinical datasets. Results are for demonstration only.

Limitations

Technical Limitations

- Ephemeral storage on free deployment tier
- No horizontal scaling
- SQLite not suitable for high-concurrency production
- No real-time wearable integration

Clinical Limitations

NOT a clinical diagnostic tool

NOT validated for medical use

- Does not account for medications, comorbidities, or family history
- Population-based guidelines may not apply to all demographics
- No clinical outcome validation

Data Limitations

- Limited to supported observation types
- Requires sufficient historical data for baseline
- Missing data handling is conservative
- No external dataset integration

Deployment

Render Free Tier Configuration

- Runtime: Python 3.14.6
- WSGI Server: Gunicorn (1 worker, 4 threads)
- Target Memory: <512 MB RAM
- Health Check: /health endpoint

Deployment Characteristics

- Stateless application design
- Ephemeral filesystem (data resets on redeploy)
- No external dependencies or paid services
- Manual backup/reset endpoints for demo data management

Local Development

- Python virtual environment
- SQLite database in instance/ directory
- Flask development server for testing

Future Work

Clinical Validation

- Validation on clinical datasets
- Collaboration with healthcare institutions
- Regulatory compliance assessment
- Clinical trial evaluation

Technical Enhancements

- PostgreSQL for production scaling
- Persistent storage layer
- Real wearable device integration
- Asynchronous task processing
- Multi-region deployment

Feature Expansion

- Additional health domains
- Medication tracking
- Comorbidity modeling
- Family history integration

Conclusion

Summary

Swasthya Twin demonstrates how Digital Twin concepts can be applied to personalized health monitoring with transparent, explainable, and reproducible computations.

Key Strengths

- Evidence-oriented design with traceable data chains
- Explainable risk assessments with feature contributions
- Scenario simulation for intervention exploration
- Reproducible and deterministic calculations
- Comprehensive security and privacy measures
- Open-source with MIT license

Resources

- GitHub: https://github.com/Sreevalli20/digital.git
- License: MIT
- Team: Brain Catalyst, VMTW

Thank you for reviewing our submission to the Happiest Health Digital Twin Challenge 2026.